

Exercise 19.2

1. Evaluate the following:

a) $\int 2x(x^2-5)^7 dx$

$$= \int (x^2-5)^7 2x dx$$

let $y = x^2 - 5$

$$\frac{dy}{dx} = 2x$$

$$dy = 2x dx$$

Now,

$$\int y^7 dy$$

$$= \frac{y^8}{8} + C$$

$$= \frac{(x^2-5)^8}{8} + C$$

b) $\int (2ax+b)(ax^2+bx+c)^n dx$

$$= \int (ax^2+bx+c)^n (2ax+b) dx$$

let, $y = ax^2 + bx + c$

$$\frac{dy}{dx} = 2ax + b$$

$$dy = (2ax + b) dx$$

Now,

$$\int y^n dy$$

$$= \frac{y^{n+1}}{n+1} + K$$

$$= \frac{1}{n+1} (ax^2+bx+c)^{n+1} + K$$

$$c) \int \frac{(x+1)}{(x^2+2x-1)^3} dx$$

$$= \int (x^2+2x-1)^{-3} (x+1) dx$$

$$\text{let, } y = x^2+2x-1$$

$$\frac{dy}{dx} = 2x+2$$

$$\frac{dy}{dx} = 2(x+1)$$

$$\frac{dy}{2} = (x+1) dx$$

Now,

$$\int y^{-3} \frac{dy}{2}$$

$$= \frac{1}{2} \int y^{-3} dy$$

$$= \frac{1}{2} \frac{y^{-3+1}}{-3+1} + C$$

$$= \frac{1}{2} \frac{y^{-2}}{-2} + C$$

$$= \frac{1}{-4(x^2+2x-1)^2} + C$$

$$d) \int \frac{x^{n-1}}{\sqrt{a^n+x^n}} dx$$

$$= \int (a^n+x^n)^{-1/2} x^{n-1} dx$$

$$\text{let } y = a^n+x^n$$

$$\frac{dy}{dx} = nx^{n-1}$$

$$\frac{dy}{n} = x^{n-1} dx$$

Now

$$\begin{aligned} & \int y^{-1/2} \frac{dy}{n} \\ &= \frac{1}{n} \int y^{-1/2} dy \\ &= \frac{1}{n} \frac{y^{-1/2+1}}{-\frac{1}{2}+1} + C \\ &= \frac{1}{n} \frac{y^{1/2}}{\frac{1}{2}} \\ &= \frac{2}{n} \sqrt{a^n + x^n} + C \end{aligned}$$

$$\begin{aligned} \text{e) } & \int x \sqrt{x^2 - 4} \, dx \\ &= \int x (x^2 - 4)^{1/2} \, dx \\ &= \int (x^2 - 4)^{1/2} x \, dx \\ & \text{let } y = x^2 - 4 \\ & \frac{dy}{dx} = 2x \\ & \frac{dy}{2} = x \, dx \end{aligned}$$

Now,

$$\begin{aligned} & \int y^{1/2} \frac{dy}{2} \\ &= \frac{1}{2} \int y^{1/2} dy \\ &= \frac{1}{2} \frac{y^{1/2+1}}{\frac{1}{2}+1} + C \end{aligned}$$

$$= \frac{1}{2} \frac{y^{3/2}}{\frac{3}{2}} + C$$

$$= \frac{1}{2} \times \frac{2}{3} (x^2 - 4)^{3/2} + C$$

$$= \frac{1}{3} (x^2 - 4)^{3/2} + C$$

$$f) \int \frac{x}{(1-x^2)^{3/2}} dx$$

$$= \int (1-x^2)^{-3/2} x dx$$

$$\text{let } y = (1-x^2)$$

$$\frac{dy}{dx} = -2x$$

$$\frac{dy}{-2} = x dx$$

Now,

$$\int y^{-3/2} \frac{dy}{-2}$$

$$= -\frac{1}{2} \int y^{-3/2} dy$$

$$= -\frac{1}{2} \frac{y^{-3/2+1}}{-\frac{3}{2}+1} + C$$

$$= -\frac{1}{2} \frac{y^{-1/2}}{-\frac{1}{2}} + C$$

$$= -\frac{1}{2} \times -\frac{2}{1} \times (1-x^2)^{-1/2} + C$$

$$= \frac{1}{\sqrt{1-x^2}} + c$$

2. Evaluate the following:

a) $\int x \cos(ax^2 + b) dx$

→ let $y = ax^2 + b$

$$\frac{dy}{dx} = 2ax$$

$$\frac{dy}{2a} = x dx$$

Now,

$$\int x \cos y dx$$

$$= \int \cos y x dx$$

$$= \int \cos y \frac{dy}{2a}$$

$$= \frac{1}{2a} \int \cos y dy$$

$$= \frac{1}{2a} \sin y + c$$

$$= \frac{1}{2a} \sin(ax^2 + b) + c$$

b) $\int \frac{1}{x} \cos(\ln x) dx$

$$= \int \cos(\ln x) \frac{1}{x} dx$$

let $y = \ln x$

$$\frac{dy}{dx} = \frac{1}{x}$$

$$dy = \frac{1}{x} dx$$

Now,

$$\int \cos(\ln x) dy$$

$$= \int \cos y dy$$

$$= \sin y + C$$

$$= \sin(\ln |x|) + C$$

$$c) \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$$

$$= \int \sin \sqrt{x} \cdot x^{-1/2} dx$$

$$\text{let } y = \sqrt{x}$$

$$y = x^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2} x^{-1/2}$$

$$2 dy = x^{-1/2} dx$$

Now,

$$\int \sin y \cdot 2 dy$$

$$= 2 \int \sin y \cdot dy$$

$$= 2x - \cos y + C$$

$$= -2 \cos \sqrt{x} + C$$

$$d) \int \sin^3 x \cos x dx$$

$$\text{let, } y = \sin x$$

$$\frac{dy}{dx} = \cos x$$

$$dy = \cos x dx$$

Now,

$$\int y^3 dy$$

$$= \frac{y^{3+1}}{3+1} + C$$

$$= \frac{\sin^4 x}{4} + C$$

e) $\int \cos^3 x \sin x \, dx$

let $y = \cos x$

$$\frac{dy}{dx} = -\sin x$$

$$-dy = \sin x \, dx$$

Now,

$$\int y^3 (-dy)$$

$$= -\int y^3 \, dy$$

$$= -\frac{y^4}{4} + C$$

$$= -\frac{\cos^4 x}{4} + C$$

f) $\int \tan^2 x \sec^2 x \, dx$

let $y = \tan x$

$$\frac{dy}{dx} = \sec^2 x$$

$$dy = \sec^2 x \, dx$$

Now,

$$\int y^2 \, dy$$

$$= \frac{y^3}{3} + C$$

$$= \frac{\tan^3 x}{3} + C$$

g) $\int \tan^2 x \sec^4 x \, dx$

let $y = \sec^2 x$

$\frac{dy}{dx} = 2 \sec^2 x \tan x$

$dy = 2 \sec^2 x \tan x \, dx$

Now,

$$\begin{aligned} & \int \tan^2 x \cdot \sec^2 x \cdot \sec^2 x \, dx \\ &= \int \tan^2 x (1 + \tan^2 x) \cdot \sec^2 x \, dx \\ &= \int y^2 (1 + y^2) \, dy \\ &= \int (y^2 + y^4) \, dy \\ &= \int y^2 \, dy + \int y^4 \, dy \\ &= \frac{y^3}{3} + \frac{y^5}{5} + C \\ &= \frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C \end{aligned}$$

h) $\int \tan^3 x \sec^{3/2} x \, dx$

let $y = \sec x$

$\frac{dy}{dx} = \sec x \cdot \tan x$

$dy = \sec x \cdot \tan x \cdot dx$

Now,

$$\begin{aligned} & \int \tan^2 x \cdot \tan x \cdot \sec x \cdot \sec^{1/2} x \, dx \\ &= \int \tan^2 x \cdot \sec^{1/2} x \cdot \sec x \cdot \tan x \, dx \\ &= \int (\sec^2 x - 1) \cdot \sec^{1/2} x \cdot \sec x \cdot \tan x \, dx \\ &= \int (y^2 - 1) y^{1/2} \cdot dy \\ &= \int (y^{5/2} - y^{1/2}) \, dy \\ &= \int y^{5/2} \, dy - \int y^{1/2} \, dy \\ &= \frac{y^{7/2}}{7/2} - \frac{y^{3/2}}{3/2} + C \\ &= \frac{2}{7} y^{7/2} - \frac{2}{3} y^{3/2} + C \end{aligned}$$

$$= \frac{2}{7} y^{7/2} - \frac{2}{3} y^{3/2} + C$$

$$= \frac{2}{7} \sec^{7/2} x - \frac{2}{3} \sec^{3/2} x + C$$

j) $\int \tan^4 x \, dx$

let $y = \tan x$

$$\frac{dy}{dx} = \sec^2 x$$

$$dy = \sec^2 x \, dx$$

Now,

$$\int \tan^2 x \cdot \tan^2 x \, dx$$

$$= \int (\sec^2 x - 1) \tan^2 x \, dx$$

$$= \int (\sec^2 x \cdot \tan^2 x - \tan^2 x) \, dx$$

$$= \int \sec^2 x \cdot \tan^2 x \, dx - \int \tan^2 x \, dx$$

$$= \int y^2 \cdot \sec^2 x \, dx - \int \tan^2 x \, dx$$

$$= \int y^2 \, dy - \int (\sec^2 x - 1) \, dx$$

$$= \int y^2 \, dy - \left[\int \sec^2 x \, dx - \int 1 \, dx \right]$$

$$= \int y^2 \, dy - \left[\int dy - \int 1 \, dx \right]$$

$$= \frac{y^3}{3} - [y - x] + C$$

$$= \frac{\tan^3 x}{3} - \tan x + x + C$$

k) $\int \cot^3 x \, dx$

let $y = \cot x$

$$\frac{dy}{dx} = -\operatorname{cosec}^2 x$$

$$-dy = \operatorname{cosec}^2 x \, dx$$

Now,

$$\begin{aligned}
 & \int \cot^3 x \, dx \\
 &= \int \cot^2 x \cdot \cot x \, dx \\
 &= \int (\operatorname{cosec}^2 x - 1) \cot x \, dx \\
 &= \int (\operatorname{cosec}^2 x \cdot \cot x - \cot x) \, dx \\
 &= \int \operatorname{cosec}^2 x \cdot \cot x \, dx - \int \cot x \, dx \\
 &= \int \cot x \cdot \operatorname{cosec}^2 x \, dx - \int \cot x \, dx \\
 &= \int y (-dy) - \int \cot x \, dx \\
 &= -\frac{y^2}{2} - \ln |\sin x| + c \\
 &= -\frac{1}{2} \cot^2 x - \ln |\sin x| + c
 \end{aligned}$$

1) $\int \operatorname{cosec}^4 x \, dx$

↳ let $y = \operatorname{cosec} x$

$$\frac{dy}{dx} = -\operatorname{cosec} x \cdot \cot x$$

$$-dy = \operatorname{cosec} x \cdot \cot x \, dx$$

Now,

$$\begin{aligned}
 & \int \operatorname{cosec}^2 x \cdot \operatorname{cosec}^2 x \, dx \\
 &= \int \operatorname{cosec}^2 x (1 + \cot^2 x) \, dx \\
 &= \int (\operatorname{cosec}^2 x + \operatorname{cosec}^2 x \cdot \cot^2 x) \, dx \\
 &= \int \operatorname{cosec}^2 x \, dx + \int \operatorname{cosec} x \cdot \cot x \cdot \operatorname{cosec} x \cdot \cot x \, dx \\
 &= \int \operatorname{cosec}^2 x \, dx + \int \operatorname{cosec} x \cdot \cot x (-dy) \\
 &= \int \operatorname{cosec}^2 x \, dx - \int \operatorname{cosec} x \cdot \cot x \, dy
 \end{aligned}$$

let $u = \cot x$

$$\Rightarrow du = -\operatorname{cosec}^2 x \, dx$$

$$\begin{aligned}
 &= -\cot x + \int u^2 \cdot (-du) \\
 &= -\cot x - \int u^2 \, du \\
 &= -\cot x - \frac{u^3}{3} + c
 \end{aligned}$$

$$= -\cot x - \frac{1}{3} \cot^3 x + c$$

$$m) \int \frac{\cos x - \sin x}{\cos x + \sin x} dx$$

$$\text{let } y = \cos x + \sin x$$

$$\frac{dy}{dx} = -\sin x + \cos x$$

$$dy = (\cos x - \sin x) dx$$

Now,

$$\int \frac{\cos x - \sin x}{\cos x + \sin x} dx$$

$$= \int \frac{1}{y} \cdot x dy$$

$$= \ln |y| + C$$

$$= \ln (\cos x + \sin x) + C$$

$$n) \int \cot x (\ln \sin x)^3 dx$$

$$\rightarrow \text{let } y = \ln \sin x$$

$$\frac{dy}{dx} = \frac{1}{\sin x} \times \frac{d(\sin x)}{dx}$$

$$dy = \frac{1}{\sin x} \times \cos x dx$$

$$dy = \cot x dx$$

Now,

$$\int \cot x (\ln \sin x)^3 dx$$

$$= \int (\ln \sin x)^3 \cot x dx$$

$$= \int y^3 dy$$

$$= \frac{y^4}{4} + C$$

$$= \frac{(\ln \sin x)^4}{4} + C$$

$$o) \int \frac{x e^n}{\cos^2(x e^n - e^n)} dx$$

$$\text{let } y = x e^n - e^n$$

$$\frac{dy}{dx} = x e^n + e^n - e^n$$

$$dy = x e^n dx$$

Now,

$$\int \frac{1}{\cos^2(x e^n - e^n)} \times x e^n dx$$

$$= \int \sec^2 y dy$$

$$= \tan y + C$$

$$= \tan(x e^n - e^n) + C$$

$$p) \int e^{\sin x \cdot \cos x} \cdot \cos 2x dx$$

$$\rightarrow \text{let } y = \sin x \cdot \cos x$$

$$\frac{dy}{dx} = \sin x \cdot \frac{d \cos x}{dx} + \cos x \cdot \frac{d \sin x}{dx}$$

$$\frac{dy}{dx} = \sin x (-\sin x) + \cos x \times \cos x$$

$$dy = (-\sin^2 x + \cos^2 x) dx$$

$$dy = \cos 2x dx$$

Now,

$$\int e^{\sin x \cdot \cos x} \cdot \cos 2x dx$$

$$= \int e^y \cdot dy$$

$$= e^y + C$$

$$= e^{\sin x \cdot \cos x} + C$$

q) $\int \frac{e^{\tan x}}{\cos^2 x} dx$

let $y = \tan x$

$\frac{dy}{dx} = \sec^2 x$

$dy = \sec^2 x dx$

Now,

$\int \frac{e^{\tan x}}{\cos^2 x} dx$

$= \int \sec^2 x e^{\tan x} dx$

$= \int e^y \cdot dy$

$= e^y + C$

$= e^{\tan x} + C$

r) $\int \left(1 - \frac{1}{x^2}\right) e^{x+1/x} dx$

let $y = x + \frac{1}{x}$

$\frac{dy}{dx} = d\left(x + \frac{1}{x}\right)$

$\Rightarrow \frac{dy}{dx} = 1 - \frac{1}{x^2} \Rightarrow dy = \left(1 - \frac{1}{x^2}\right) dx$

Now,

$\int e^{x+1/x} \left(1 - \frac{1}{x^2}\right) dx$

$= \int e^y dy$

$= e^y + C$

$= e^{x+\frac{1}{x}} + C$

$e^{x+\frac{1}{x}} + C$

$$5) \int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}} dx$$

$$\text{let } y = \sin^{-1}x$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$dy = \frac{1}{\sqrt{1-x^2}} dx$$

Now,

$$\int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}} dx$$

$$= \int e^y dy$$

$$= e^y + C$$

$$= e^{\sin^{-1}x} + C$$

3) Evaluate the following

$$a) \int \frac{e^x}{1+e^x} dx$$

$$\text{let } y = 1+e^x$$

$$\frac{dy}{dx} = 0 + e^x$$

$$dy = e^x dx$$

Now,

$$\int \frac{1}{1+e^x} e^x dx$$

$$= \int \frac{1}{y} dy$$

$$= \ln|y| + C$$

$$= \ln|1+e^x| + C$$

$$b) \int \frac{e^{2n}}{1+e^n} dn$$

$$\text{let } y = 1+e^n \Rightarrow e^n = y-1$$

$$\frac{dy}{dn} = 0+e^n$$

$$dy = e^n dn$$

Now,

$$\int \frac{e^{2n}}{1+e^n} dn$$

$$= \int \frac{e^n \times e^n}{1+e^n} dn$$

$$= \int \frac{e^n}{1+e^n} dy$$

$$= \int \frac{e^n}{y} dy$$

$$= \int \frac{y-1}{y} dy$$

$$= \int 1 - \frac{1}{y} dy$$

$$= \int 1 dy - \int \frac{1}{y} dy$$

$$= y - \ln|y| + c$$

$$= 1+e^n - \ln|1+e^n| + c$$

$$c) \int \frac{e^n - 1}{e^n + 1} dn$$

$$= \text{let } y = e^n + 1$$

$$\frac{dy}{dn} = e^n$$

$$dy = e^n dn$$

Now,

$$\int \frac{e^n - 1}{e^n + 1} dn$$

$$= \int \frac{e^n - 1}{y} dn$$

$$= \int \frac{e^n}{y} - \frac{1}{y} dn$$

$$= \int \frac{e^n}{y} dn - \int \frac{1}{y} dn$$

$$= \int \frac{1}{y} dy - \int \frac{1}{y} dn$$

$$= \ln|y| - \int \frac{y - e^n}{y} dn$$

$$= \ln|y| - \int \left(1 - \frac{e^n}{y}\right) dn$$

$$= \ln|y| - \left[\int 1 dn - \int \frac{e^n}{y} dn \right]$$

$$= \ln|y| - \left[n - \int \frac{1}{y} dy \right]$$

$$= \ln|y| - [n - \ln|y|] + c$$

$$= \ln|y| - n + \ln|y| + c$$

$$= 2\ln|y| - n + c$$

$$= 2\ln|e^n + 1| - n + c$$

4) Evaluate the following definite integral.

$$a) \int_0^1 x \sqrt{x^2+1} dx$$

$$= \int_0^1 x \cdot (x^2+1)^{1/2} dx$$

$$\text{let } y = x^2 + 1$$

$$\frac{dy}{dx} = 2x$$

$$dy = 2x dx$$

$$\frac{dy}{2} = x dx$$

$$\text{When } x=0$$

$$y = 1$$

$$\text{When } x=1$$

$$y = 2$$

Now,

$$\int_1^2 x \cdot y^{1/2} \frac{dy}{2}$$

$$= \frac{1}{2} \left[\frac{y^{1/2+1}}{\frac{1}{2}+1} \right]_1^2$$

$$= \frac{1}{2} \left[\frac{y^{3/2}}{\frac{3}{2}} \right]_1^2$$

$$= \left[\frac{2}{3} y^{3/2} \right]_1^2 \times \frac{1}{2}$$

$$= \frac{1}{2} \times \frac{2}{3} \left[2^{3/2} - 1^{3/2} \right]$$

$$= \frac{1}{3} \left[2\sqrt{2} - 1 \right]$$

$$b) \int_0^a \frac{x}{\sqrt{a^2+x^2}} dx$$

$$= \int_0^a (a^2+x^2)^{-1/2} x dx$$

let $y = a^2+x^2$

When $x=0$

$$\frac{dy}{dx} = 2x$$

$y = a^2$

When $x=a$

$$dy = 2x dx$$

$y = 2a^2$

$$\frac{dy}{2} = x dx$$

Now

$$\int_{a^2}^{2a^2} y^{-1/2} \frac{dy}{2}$$

$$= \frac{1}{2} \left[\int y^{-1/2} dy \right]_{a^2}^{2a^2}$$

$$= \frac{1}{2} \times \left[\frac{y^{-1/2+1}}{-\frac{1}{2}+1} \right]_{a^2}^{2a^2}$$

$$= \frac{1}{2} \left[\frac{y^{1/2}}{\frac{1}{2}} \right]_{a^2}^{2a^2}$$

$$= \frac{1}{2} \times \frac{2}{1} \left[(2a^2)^{1/2} - (a^2)^{1/2} \right]$$

$$= (\sqrt{2}a)^{1/2} - a$$

$$= \sqrt{2}a - a$$

$$= a(\sqrt{2}-1)$$

$$c) \int_0^{\pi/4} e^{\cos 2x} \cdot \sin x \cdot \cos x \, dx$$

$$\rightarrow \text{let } y = \cos 2x$$

$$\frac{dy}{dx} = -\sin 2x$$

$$dy = -\sin 2x \cdot \cos x \, dx$$

$$-\frac{dy}{2} = \sin x \cdot \cos x \, dx$$

$$\text{When } x=0 \Rightarrow y=1$$

$$\text{When } x=\frac{\pi}{4} \Rightarrow y=0$$

Now,

$$\int_0^{\pi/4} e^{\cos 2x} \cdot \sin x \cdot \cos x \, dx$$

$$= \int_1^0 e^y \left(-\frac{dy}{2} \right)$$

$$= -\frac{1}{2} [e^y]_1^0$$

$$= -\frac{1}{2} [e^0 - e^1]$$

$$d) \int_0^{\pi/2} \frac{\sin x}{(1+\cos x)^3} dx$$

$$= \int_0^{\pi/2} \sin x (1+\cos x)^{-3} dx$$

$$\text{let } y = 1 + \cos x$$

$$\text{When } x=0$$

$$\frac{dy}{dx} = 0 + (-\sin x)$$

$$y = 2$$

$$\text{When } x = \pi/2$$

$$dy = -\sin x dx$$

$$y = 1$$

$$-dy = \sin x dx$$

Now,

$$\int_2^1 y^{-3} (-dy)$$

$$= - \left[\frac{y^{-3+1}}{-3+1} \right]_2^1$$

$$= - \left[\frac{y^{-2}}{-2} \right]_2^1$$

$$= +\frac{1}{2} [1^{-2} - 2^{-2}]$$

$$= +\frac{1}{2} \left[\frac{1}{1} - \frac{1}{4} \right]$$

$$= +\frac{1}{2} \left[\frac{3}{4} \right]$$

$$= +\frac{3}{8}$$

Ans

$$c) \int_0^{\pi/4} \cos^3 x \cdot \sin^2 x \, dx$$

→ let $y = \sin x$ when $x=0$

$$\frac{dy}{dx} = \cos x$$

$$y = 0$$

when $x = \pi/4$

$$dy = \cos x \, dx$$

$$y = \frac{1}{\sqrt{2}}$$

Now,

$$\int_0^{\pi/4} \cos^2 x \cdot \sin^2 x \cdot \cos x \, dx$$

$$= \int_0^{\pi/4} (1 - \sin^2 x) \cdot \sin^2 x \cdot \cos x \, dx$$

$$= \int_0^{1/\sqrt{2}} (1 - y^2) y^2 \, dy$$

$$= \int_0^{1/\sqrt{2}} (y^2 - y^4) \, dy$$

$$= \left[\int y^2 \, dy \right]_0^{1/\sqrt{2}} - \left[\int y^4 \, dy \right]_0^{1/\sqrt{2}}$$

$$= \left[\frac{y^3}{3} \right]_0^{1/\sqrt{2}} - \left[\frac{y^5}{5} \right]_0^{1/\sqrt{2}}$$

$$= \left(\frac{1}{\sqrt{2}} \right)^3 - 0 - \left(\frac{1}{\sqrt{2}} \right)^5 + 0$$

$$= \frac{1}{3} \times \left(\frac{1}{\sqrt{2}} \right)^3 - \frac{1}{5} \times \left(\frac{1}{\sqrt{2}} \right)^5$$

$$= \frac{1}{3} \times \frac{1}{2\sqrt{2}} - \frac{1}{5} \times \frac{1}{4\sqrt{2}}$$

$$= \frac{1}{6\sqrt{2}} - \frac{1}{20\sqrt{2}}$$

$$= \frac{14}{\sqrt{2}}$$

$$f) \int_0^{\pi/2} (1 + \cos x)^2 \cdot \sin x \, dx$$

$$= \int_0^{\pi/2} \sin x (1 + \cos x)^2 \, dx$$

let $y = 1 + \cos x$ When $x = 0$
~~1~~ $x = 0$

$$\frac{dy}{dx} = 0 - \sin x$$

$$y = 2$$

When $x = \pi/2$

$$-dy = \sin x \, dx$$

$$y = 1$$

Now,

$$\int_2^1 y^2 (-dy)$$

$$= - \left[- \int y^2 \, dy \right]_2^1$$

$$= - \left[\frac{y^3}{3} \right]_2^1$$

$$= - \frac{1}{3} [1^3 - 2^3]$$

$$= - \frac{1}{3} [1 - 8]$$

$$= - \frac{1}{3} \times -7$$

$$= \frac{7}{3}$$

2

$$g) \int_0^{\pi/4} \tan^3 x \, dx$$

$$= \int_0^{\pi/4} \tan^2 x \cdot \tan x \, dx$$

let $y = \tan x$

When $x = 0$

$$\frac{dy}{dx} = \sec^2 x$$

$$y = 0$$

$$dy = \sec^2 x \, dx \quad \text{When } y = \pi/4$$

Now,

$$y = 1$$

$$\int_0^{\pi/4} (-1 + \sec^2 x) \, dx$$

$$= \int_0^{\pi/4} (-\tan x + \tan x \cdot \sec^2 x) \, dx$$

$$= \left[-\int \tan x \, dx \right]_0^{\pi/4} + \left[\int \tan x \cdot \sec^2 x \, dx \right]_0^{\pi/4}$$

$$= \left[-\int \tan x \, dx \right]_0^{\pi/4} + \left[y \cdot dy \right]_0^1$$

$$= \left[-\int \tan x \, dx \right]_0^{\pi/4} + \frac{1}{2} [y^2]_0^1$$

$$= \left[-\int \tan x \, dx \right]_0^{\pi/4} + \frac{1}{2} [1^2 - 0^2]$$

$$= \left[-\ln(\sec x) \right]_0^{\pi/4} + \frac{1}{2}$$

$$= -[\ln(\sec \pi/4) - \ln(\sec 0)] + \frac{1}{2}$$

$$= -[\ln(\sqrt{2}) - \ln(1)] + \frac{1}{2}$$

$$= -\frac{1}{2} \ln 2 + \frac{1}{2}$$

$$= \frac{1}{2} - \frac{1}{2} \ln 2$$

$$\text{h) } \int_1^2 e^{2x^2-1} \cdot 2x \, dx$$

$$\rightarrow \text{let } y = 2x^2 - 1 \quad \text{When } x = 2$$

$$\frac{dy}{dx} = 4x \quad y = 7$$

$$dy = 4x \, dx \quad \text{When } x = 1$$

$$y = 1$$

$$\frac{dy}{2} = 2x dx$$

∴ Now,

$$\int_1^2 2x e^{2x^2-1} dx$$

$$= \int_1^2 e^{2x^2-1} 2x dx$$

$$= \int_1^7 e^y \frac{dy}{2}$$

$$= \frac{1}{2} [\int e^y dy]_1^7$$

$$= \frac{1}{2} [e^y]_1^7$$

$$= \frac{1}{2} [e^7 - e^1]$$

$$= \frac{1}{2} e [e^6 - 1]$$

∴